PGCP-AI August 2026
C-DAC THIRUVANANTHAPURAM

Java Programming– LAB 4

1. Write an interface called **Result** with an abstract method **checkStatus(int mark)** that returns a **boolean** value. Write a class called **Exam** which implements both Result and Classify. The checkStatus() method should return true if the mark is greater than or equal to 50 else false. Write another interface called **Classify** with an abstract method **checkGrade(int average)** which returns a **string**. The checkGrade() method must return "First class" when the parameter average is 60 or more, " Second class" when average is 50 or more but below 60, "Third class" when average is less than 50. Now create an object for Exam class and call their respective methods.

2. Create one interface named **Sample** which contains a variable named **x,** one abstract method **square()** to find the square of the number, a default method **cube()** to find the cube of the number and a static method **meth1()** to print the value of the variable. Write the implementing class **SampleMain** and invoke the static method using interface reference and invoke the other two methods using class object.

3. Create a functional interface named **Verify** with one abstract method **boolean check(int a)**. Create a class which invokes the method using lambda expression where the method returns true if the number is prime and false otherwise.
4. Create a functional interface named **Concatenation** with one abstract method **String join(String a, String b)**. Write a program to implement lambda expression to concatenate two strings.
5. Write a program to create two child threads using Thread class, one to compute the first 25 prime numbers and the other to compute the first 50 fibonacci numbers. After calculating 25 fibonacci numbers, make that thread to sleep and take up prime number computation.
6. Write a Java program to create two child threads using Runnable interface, one to find the sum of natural numbers from 1 to n and the other to display the factorial of the numbers upto n.
7. Create a child thread to print Right Triangle star pattern using Lambda expression.

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